Project Management in Infrastructure

- Project planning and scheduling are fundamental in civil engineering to ensure infrastructure projects are completed on time, within budget, and with optimal resource use.
- Two key techniques are the Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT).
- These network-based methods help identify activity sequences, durations, and critical tasks that determine the project's overall timeline.

Importance of Planning and Scheduling

- Minimizes cost overruns
- Reduces delays
- Ensures optimal resource utilization
- Enhances coordination between stakeholders
- Improves project quality & safety

CPM is a deterministic project scheduling method that uses fixed activity durations to determine the longest path in a project network.

Basic Terms

- Activity
- Event
- Duration
- Early Start (ES) / Early Finish (EF)
- Late Start (LS) / Late Finish (LF)
- Float / Slack

Types of Floats

- Total Float
- Free Float
- Independent Float
- Interfering Float

Steps in CPM

1. Identify activities
2. Establish relationships (precedence)
3. Draw network diagram (AON is common)
4. Estimate activity durations
5. Forward pass calculation (ES/EF)
6. Backward pass calculation (LS/LF)
7. Determine critical path (zero float)

Crashing & Time-Cost Trade-off

Used to accelerate project completion by allocating additional resources or overtime.

Crashing Concepts

- Cost slope
- Direct vs indirect costs
- Crash limit
- Optimal time-cost curve

PROGRAM EVALUATION & REVIEW TECHNIQUE (PERT)

PERT is used for projects with uncertain activity durations, especially R&D, defense, and complex infrastructure projects.

Expected Duration (te)

$$t_e = \frac{O + 4M + P}{6}$$

Variance

$$\sigma^2 = \left(\frac{P - O}{6}\right)^2$$

3.3 Probability Calculation

Using Z-values:

$$Z = \frac{T - \mu}{\sigma}$$

Where T = target time, μ = expected project time, σ = standard deviation.

Time Estimates

- Optimistic time (O)
- Most likely time (M)
- Pessimistic time (P)

Applications

- Risk analysis
- Uncertain projects
- Civil infrastructures with variable weather and environmental conditions

RESOURCE ALLOCATION & MANAGEMENT

Types of Resources

- Manpower (labor, engineers)
- Materials (cement, steel, aggregates)
- Machinery (cranes, excavators)
- Money
- Method & technology

Resource Allocation Issues

- Overallocation
- Idleness
- Underutilization
- Peak demand

Resource Levelling

Objective: Minimize fluctuations in resource usage.

Techniques:

- Shifting non-critical activities
- Using float
- Smoothing manpower curves

Resource Allocation (When resources are limited)

Methods:

- Minimum Slack Priority
- Shortest Processing Time
- Earliest Due Date

Modern Tools

- Primavera P6
- MS Project
- BIM & 4D Scheduling
- Lean Construction (Last Planner System)

CONTRACT MANAGEMENT AND LEGAL ASPECTS

Types of Construction Contracts

1. **Lump Sum Contract**
2. **Item Rate Contract**
3. **Cost Plus Contract**
4. **Turnkey Contract**
5. **BOT/BOOT/PPP Contracts**
6. **EPC Contracts**

Legal Aspects

- Tendering laws
- Contract Act principles (Offer, Acceptance, Consideration)
- Obligations & liabilities
- Liquidated damages (LD)
- Arbitration & Dispute Resolution
- FIDIC norms
- Environmental & safety regulations

Contract Documents

- General Conditions of Contract (GCC)
- Special Conditions of Contract (SCC)
- Technical Specifications
- Bill of Quantities (BOQ)
- Drawings
- Tender documents

Claims & Variations

- Time extension claims
- Cost escalation
- Delays (Excusable, Non-Excusable, Compensable)

QUALITY CONTROL & QUALITY ASSURANCE (QC/QA)

Difference Between QC and QA

- **Quality Control:** Operational techniques to verify quality (tests, inspections).
- **Quality Assurance:** Systematic processes ensuring quality (procedures, SOPs, audits).

QC in Construction

- Soil testing
- Concrete tests: slump test, cube strength
- Steel tests: tensile test, bend test
- Asphalt tests
- NDT techniques (Ultrasonic Pulse Velocity, Rebound Hammer)

QA Methods

- Material approval system
- Method statements
- Checklists
- Inspection & Test Plans (ITPs)
- ISO 9001 standards
- QA documentation

Total Quality Management (TQM)

- Continuous improvement (Kaizen)
- Six Sigma approach
- Quality circles